Abstract Algebra - Stockholm University 2025

1. Groups

1.1. Conjugation, centralizer, center, normalizer

For $g \in G$, the conjugation map $c_g(x) = gxg^{-1}$ is an automorphism of G (an *inner* automorphism). For a subset $S \subseteq G$, the centralizer is

$$C_G(S) = \{\, g \in G : \ gs = sg \ \text{ for all } s \in S \,\}; \quad C_G(g) := C_G(\{g\}).$$

The center is

$$Z(G) = C_G(G) = \{g \in G : \ gx = xg \ \forall x \in G\}$$

(always a characteristic, hence normal, subgroup). For a subgroup $H \leq G$, the normalizer is

$$N_G(H) = \{\, g \in G : \ gHg^{-1} = H \,\},$$

the largest subgroup of G in which H is normal (so $H \trianglelefteq N_G(H)$). One always has $C_G(H) \leq N_G(H)$. The number of conjugates of H in G equals $[G : N_G(H)]$.

1.2. Normal subgroups and quotients

A subgroup $N \leq G$ is **normal** (write $N \trianglelefteq G$) if $gNg^{-1} = N$ for all $g \in G$. Equivalently, every left coset equals the corresponding right coset: $gN = Ng$. If $N \trianglelefteq G$, the set of cosets $G/N = \{gN : g \in G\}$ is a group with $(gN)(hN) = (gh)N$.

1.3. Isomorphism theorems

(1) **First isomorphism theorem.** For $\varphi : G \to H$ a homomorphism,

$$G/\ker\varphi \ \cong \ \operatorname{im}\varphi.$$

(2) **Second isomorphism theorem.** For $A \leq G$ and $N \trianglelefteq G$,

$$A \cap N \trianglelefteq A, \qquad AN \leq G, \qquad AN/N \ \cong \ A/(A \cap N).$$

(3) **Third isomorphism theorem.** If $K \leq H \trianglelefteq G$, then $H/K \trianglelefteq G/K$ and

$$(G/K)/(H/K) \ \cong \ G/H.$$

(4) **Correspondence (lattice) theorem.** For $N \trianglelefteq G$, the map

$$\{\, \text{subgroups } H \mid N \leq H \leq G \,\} \ \longleftrightarrow \ \{\, \text{subgroups of } G/N \,\}, \quad H \mapsto H/N$$

is a bijection preserving inclusion and index; it carries normal subgroups $H \trianglelefteq G$ with $N \leq H$ to normal subgroups of G/N.

2. Group actions

3. Sylows Theorem

4. Rings

5. Integral -, Unique Factorization - & Principle Ideal Domains

6. Field Extensions